

GPU Needs & Search Architecture

How we match industry multimodal retrieval — compute plan, architecture cutover, and search quality path

System: DriveFaceIndexer (video / image search + Carousel Studio)
Date: 2026-09-02 · **Audience:** product, ops, engineering
Status: Expanded architecture brief for GPU provisioning and search roadmap

Decision. Buy **GPU for InsightFace** (CUDA) first; optional local VLM later. Keep Railway for API/indexer/Postgres/Qdrant/Drive. Keep Gemini/Claude as APIs — not a face-GPU substitute. Default / search stays until golden eval says promote.

1. Why this brief exists

Three pressures: **(1)** people-index throughput — InsightFace on CPU is the face saturator and Railway has no GPU; **(2)** search relevance — visual ANN is coarse recall, so apparel/brand and gym lookalikes feel meaningless when ANN is treated as precision; **(3)** architecture cutover — split API / indexer / face-GPU while adding OCR geometry and a budgeted top-K judge without harming live search.

2. What other companies actually do

Mature visual search (Lens-class, Pinterest, Shopify visual search, multimodal RAG) is almost never “embed the query → return nearest neighbors.” ANN is only **coarse recall**. Precision comes later.

2.1 Industry stack — retrieve then rerank

Stage	What happens	Cost	Role
A. Cheap retrieve	Bi-encoder ANN (CLIP / SigLIP / multimodal embed) pulls hundreds of candidates	Low	Recall prior only
B. Structure	OCR boxes, objects/apparel, face IDs, metadata, lexical fuse	Medium	Disambiguate what ANN cannot
C. Expensive judge	Cross-encoder or VLM scores query ↔ candidate on top ~20–100	High	Precision
D. Product policy	Hard negatives, diversity, person/role gates, brand-on-garment rules	Low	Product truth

Needing a VLM on a visually similar shortlist is **normal**. Shipping ANN alone would look equally “meaningless” for “logo on a tee vs logo on a banner.” Embeddings are a prior; they are not a judge.

2.2 What embeddings can and cannot do

Embeddings are good at	Embeddings are bad at
Pulling gym / campus / party-ish lookalikes	Identity of near-duplicate objects (sandbag vs medicine ball; ski-erg vs rower)
Broad scene neighborhood	Brand location (tee print vs backdrop)
Cheap corpus-wide recall	Yes/no on a specific natural-language claim

2.3 DriveFaceIndexer mapping

Industry block	Our equivalent today	Gap / next move
Visual ANN recall	Qdrant + Gemini embedding-2	Keep as recall; demote as sole ranker
Caption / text index	Caption ANN + lexical + stored captions	Caption-first fusion prototyped; not default
Object / attribute gates	Object labels + apparel-brand association + hard-negatives	Backdrop brand still co-mentioned in captions
OCR localization	RapidOCR ONNX lane (opt-in, idle workers)	Region heuristics incomplete (many spans → other)
Face identity	InsightFace buffalo_I on CPU ONNX	Move to CUDA GPU workers
Top-K multimodal judge	Caption LLM filter / Gemini VLM paths	True pixel yes/no on shortlist
Eval / regression SLO	eval script + small golden; caption_queries_500 unwired	Wire as deploy / nightly gate (B8)

2.4 Failures only structure + judge fix

Failure	Why ANN / captions fail	What fixes it
Brand on backdrop ranked as on-garment	Caption says both "hat" and brand name	OCR boxes → brand near torso/garment, not banner
Sandbag vs medicine ball	Shape neighbors in embed space	Better object tags and/or top-K pixel yes/no
Ski-erg vs rower	Same	Same
Missing faces in people search	CPU InsightFace backlog / incomplete index	GPU face fleet + coverage SLO

3. Current system baseline

3.1 Topology

Users → Indexer UI / Carousel

- dfi-backend (API; historically coupled to index work)
- dfi-face-worker (CPU InsightFace + idle OCR/object lanes)
- Postgres (SoR) · Qdrant (vectors) · /app/data volume
- Drive + Gemini + Claude / OpenRouter

Documented same-image split: dfi-backend (API) · dfi-indexer (volume, downloads, Whisper, ffmpeg) · dfi-face-worker fleet (volume-less on Railway today). See docs/indexer-service.md.

3.2 Search pipeline (images)

Query → cache → persons/roles → route (video | person | object/apparel/scene)
 → local DB + Qdrant ANN (visual + caption + objects)
 → captions → object/apparel/action gates → optional LLM/VLM → ranked files

Quality leaks: weak captions · mode collisions · visual ANN ≠ identity · backdrop brand co-mention · cache without quality version · no deploy-time search SLO.

3.3 Face path today

- Model: InsightFace buffalo_l · Providers default: CPUExecutionProvider only
- Jobs: durable face_jobs with FOR UPDATE SKIP LOCKED when FACE_JOBS_ENABLED
- Workers: Railway CPU (~2 GB / 2 vCPU), sequential InsightFace lock
- OCR: RapidOCR ONNX idle lane; ocr_lane_enabled defaults false; no Drive download on search path

4. GPU needs — purchase brief

4.1 Workload → compute ownership

Workload	Where it should run	Buy / fund
API, search, OAuth, Range streaming	Railway CPU (dfi-backend)	Keep Railway
Downloads, Whisper, ffmpeg, enqueue	Railway indexer + volume	Same-image indexer split
InsightFace / ArcFace	GPU host (CUDAExecutionProvider)	RunPod / Modal / Vast / GCP/AWS GPU
Optional local VLM (Qwen path wired, off)	GPU sidecar (phase 2)	Only after face GPU healthy
RapidOCR ONNX	CPU idle lane on face workers	No GPU required
Gemini captions / embeds / VLM API	Google API	Paid TPM — not a GPU box
Carousel scripts	Anthropic Claude (+ OpenRouter)	API credits
Postgres + Qdrant	Railway / managed data	Keep

4.2 Recommended starting SKU

Item	Recommendation
GPU class	1× NVIDIA L4 or T4-class, 16–24 GB VRAM
Providers	RunPod · Modal · Vast.ai · GCP L4 · AWS g4dn / L4
Process	Same or thin face-worker image; claim face_jobs from Postgres
Env	INSIGHTFACE_PROVIDERS=[CUDAExecutionProvider, CPUExecutionProvider]
Concurrency	Start 1–2 parallel jobs; raise only with VRAM headroom
Storage	No shared Drive volume; short-lived cache / re-fetch bytes
HA	1 GPU worker first; add second when backlog SLO fails

Do not put on the GPU box: user-facing API, authoritative Postgres/Qdrant, long-lived Drive cache as SoR, or unbounded GGUF VLM that starves InsightFace.

4.3 Phase-2 GPU (optional local judge)

Only if API VLM cost/latency becomes the limiter after structure lands: Qwen3-VL sidecar (already stubbed), or MiniCPM/Moondream yes/no on top-20. A mini-VLM that re-captions the whole image will **re-learn backdrop mistakes** — prefer OCR geometry first.

4.4 Three wallets

- **Platform (Railway):** API, indexer, volume, data adjacency.
- **GPU (external):** InsightFace first; optional local VLM later.
- **Models (APIs):** Gemini / Claude / OpenRouter. Raise TPM only after interactive vs backfill isolation.

5. Target architecture

5.1 Target topology

Users → Frontends

- API service [Railway CPU] search, auth, streaming
- Indexer service [Railway + volume] downloads, whisper, ffmpeg, enqueue
- Face GPU fleet [external CUDA] InsightFace; later optional VLM
- Enrichment lanes [CPU] RapidOCR, objects – yield to faces

Shared: Postgres (SoR) · Qdrant (derived) · media lifecycle

5.2 Search path we are moving toward

- Retrieve → visual + caption + object candidates (existing indexes)
- Fuse → caption/object-first; visual ANN as recall floor
- Structure → apparel association · OCR torso/upper vs signage · hard-negatives
- Judge → budgeted VLM / caption LLM on top-N only
- Promote → only after golden eval beats production default

5.3 Isolation rules (non-negotiable)

- Do not change default production ranking until eval says so.
- OCR / experimental fusion must not Drive-download or commit the request DB session on the search path.
- GPU face service: health + claim semantics; API never blocks on InsightFace.
- Gemini interactive quota isolated from background caption/embed backfill.
- OCR lane and backfill default off; enable only for controlled enrich windows.

6. Implementation roadmap

Phase	Work	Outcome	Compute
0 Stabilize	API/indexer isolation; disk auto-clean; healthy face_jobs	Index finishes; UI stays live	Railway CPU
1 GPU faces	External CUDA workers; CUDA-first providers	People coverage & speed	GPU SKU
2 Structure	OCR regions; apparel association; object hard-negatives	Brand-on-garment ≠ backdrop	CPU ONNX
3 Rank policy	Promote caption/structure fusion after A/B + golden	Less lookalike junk in top-5	No new GPU
4 Top-K judge	Budgeted VLM rerank on top 20–50 hard queries	Industry-grade precision	API or GPU VLM
5 SLO	Golden + caption_queries_500 on deploy; typed slices	No silent regressions	Existing Gemini

6.1 Phase acceptance checklists

Phase 1 — GPU faces

- Face worker reports CUDA provider in use
- face_jobs backlog drains under load without API latency spike
- People-query precision@5 holds or improves on fixed set
- Failover to CPU only if CUDA unavailable (alert)
- No volume mount required for horizontal GPU scale

Phase 2–3 — Structure + rank

- OCR spans carry usable torso / upper / signage regions (not ~all other)
- Apparel-brand A/B: backdrop demoted on hard queries
- Object hard-negatives still pass (rowing / sandbag / ski-erg)
- Production /search unchanged until promote decision

Phase 4–5 — Judge + SLO

- VLM judge only on shortlist; spend per query capped
- eval_search_quality (+ 500-query set) on deploy or nightly
- Separate scores: people · scene · apparel_brand · object · OCR/text

7. Implementing what companies do — concrete moves

Industry practice	Our implementation move
ANN as recall only	Keep Qdrant visual ANN; demote fusion weight; strong visual floor for missing-caption fallback
Multi-signal retrieve	Caption ANN + lexical + object DB already hybrid; keep
OCR for text-in-image	RapidOCR on idle workers; persist spans; association uses region
Face identity	InsightFace on GPU; face tags in person queries
Cross-encoder / VLM rerank	Caption LLM now; true pixel yes/no on top-K next
Offline eval + regression	Wire golden + caption_queries_500; make B8 a real gate
Service isolation	API ≠ indexer ≠ face GPU (same pattern as large media platforms)

Order that actually moves quality

1. GPU InsightFace (coverage + people latency)
2. OCR geometry into apparel-brand association
3. Promote structured ranking only with measured A/B
4. Top-K VLM for residual lookalikes OCR cannot settle
5. Deploy SLO so we stop whack-a-mole

GGUF-first on Railway face workers is the romantic path. **CUDA faces + OCR-on-ONNX** match measured failures and the existing stack.

8. Success metrics

Metric	Signal we care about
Face job latency / backlog	GPU workers drain queue without starving API
People precision@5	Named faces surface before lookalikes
Apparel-brand precision@5	On-garment hits over backdrop/signage
Object / action precision@5	Hard negatives (rowing, ski-erg, sandbag) hold
VLM judge spend / query	Only top-K judged; no full-corpus VLM
Golden / 500-query on deploy	Fail closed or alert on regression
Caption/index completeness	Embeds + captions present for PROCESSED media

9. Risks and mitigations

Risk	Mitigation
GPU vendor lock / spot eviction	CUDA + CPU fallback providers; job lease/retry already designed
Egress cost GPU ↔ Postgres/Drive	Batch claims; short-lived image cache on GPU host
Search experiment harms prod	Isolated policy/endpoint; OCR lane off by default
OCR mis-region → wrong demotion	Require confidence + region; A/B before promote
VLM spend explosion	Hard top-K + concurrency cap; starve if face backlog
Quota 429s during backfill	Interactive vs backfill isolation (quality blocker B5)

10. Bottom line

Other companies do **not** ship “embeddings = answer.” They ship **retrieve** → **structure** → **top-K judge**, with faces and OCR as first-class signals. We align the same way: **GPU for InsightFace**, Railway for serve/index, structure (OCR/objects/captions) before trusting rank, judge only on a shortlist, and promote only with eval — production search stays safe until then.

Appendix A — Config knobs

```
# Face / GPU
INSIGHTFACE_PROVIDERS=["CUDAExecutionProvider", "CPUExecutionProvider"]
FACE_JOBS_ENABLED=true  RUN_FACE_WORKER=true  FACE_WORKER_CONCURRENCY=1

# API / Indexer
API: RUN_INDEXER=false  RUN_FACE_WORKER=false
Indexer: RUN_INDEXER=true  FACE_JOBS_ENABLED=true

# OCR (opt-in runtime)  ocr_lane_enabled=false  ocr_backfill_enabled=false
# Local VLM (later)    QWEN_VLM_ENABLED=false  QWEN_VLM_BASE_URL=http://...
```

Appendix B — In-repo reading

- docs/indexer-service.md — API / indexer / face-worker split
- docs/quality-blockers-and-subscriptions.md — B1–B8 quality ladder
- docs/full-system-architecture-and-risks.md — full topology and risks
- backend/scripts/eval_search_quality.py — golden judge
- datasets/caption_queries_500.csv — broader eval substrate (wire next)

Markdown source (editable twin): docs/gpu-needs-and-search-architecture.md